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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Machine Learning-Based Email Spam Detection and Intelligent Email Classification System. The internship spanned an 8-week period and was undertaken to apply Natural Language Processing (NLP) and machine learning techniques to cybersecurity challenges. The primary objective of this internship was to gain proficiency in data analysis, text processing, machine learning algorithms, and software development to enhance employability skills while solving a critical communication security problem.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning system using Python, Pandas, and Scikit-learn that can accurately classify emails into spam and legitimate categories.
- To integrate Natural Language Processing (NLP) techniques for extracting meaningful features from unstructured email content, including subject lines, body text, and sender information.
- To implement interactive data visualizations that help administrators understand spam patterns, feature correlations, and model performance metrics.
- To evaluate and compare different classification algorithms (Logistic Regression, Random Forest, Gradient Boosting, Naive Bayes) to find the most effective model for spam detection.
- To design a scalable system architecture that can process large volumes of incoming emails in real-time while maintaining high detection accuracy.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational predictive model capable of classifying emails with high accuracy and an F1-score of 0.48 using Gaussian Naive Bayes on the extracted feature set.

- Organizations can now automatically filter malicious content, reducing security risks associated with phishing attacks and improving overall communication efficiency.

- Comprehensive data visualizations including class distribution charts, feature importance plots, and confusion matrices that enhance administrative monitoring capabilities.

- A robust feature engineering pipeline that successfully extracts over 30 distinct attributes from raw email data, including sender reputation, URL counts, and suspicious keyword frequencies.

- The classification system can be extended with advanced features such as deep learning models (LSTMs or Transformers) for more nuanced text understanding and continuous online learning.

These outcomes directly address the problem statement by providing a modern and intelligent email security solution that improves spam detection accuracy, enhances communication security, reduces unwanted emails, and protects users from malicious content.

CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a leading technology solutions provider focused on bridging the academia-industry divide, enhancing student employability, promoting innovation, and fostering an entrepreneurial ecosystem in the cybersecurity sector. By leveraging emerging technologies such as Artificial Intelligence and Machine Learning, the organization aims to augment and upgrade the digital security ecosystem, enabling enterprises to protect their critical communication infrastructure.

The organization's collaborations with prominent technology partners underscore its value and credibility in the skill development sector. Through projects like the Machine Learning-Based Email Spam Detection System, the organization demonstrates its commitment to applying cutting-edge technology to solve pressing security challenges.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful cybersecurity solutions to drive digital safety and enterprise resilience.

- **Mission:** To support organizations dedicated to protecting user data by empowering and equipping IT administrators with intelligent threat detection tools, thereby creating a secure digital communication network.

- **Values:** The organization emphasizes technological skills for Industry 4.0, data privacy, ethical AI development, and inclusive access to secure communication for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, especially sensitive email datasets and security protocols used in threat modeling.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members and mentors.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share innovative ideas regarding NLP applications, and contribute to the organization's goals.
- **Compliance:** Interns must comply with all organizational policies, including ethical guidelines for AI development and data usage.

2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight for cybersecurity initiatives.
- **Executive Director:** Oversees day-to-day operations and implementation of security programs.
- **Project Managers:** Lead specific initiatives such as the development of threat detection software and AI tools.
- **Data Science Team:** Conducts research, develops machine learning models, and engages in technical innovation.

- **Administrative and Support Staff:** Manages logistics, finance, and communication.

- **Interns:** Work under the guidance of project managers and data scientists to contribute to ongoing technical projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees guiding the intern include:

1. Project Managers:
    - Design and implement technical projects.
    - Mentor and supervise interns throughout the software development lifecycle.
    - Coordinate with enterprise stakeholders to gather security requirements.
2. Senior Data Scientists:
    - Provide technical guidance on NLP algorithms and feature engineering.
    - Review code and evaluate classification performance metrics.
    - Assist in troubleshooting technical issues during implementation.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

Email remains one of the most widely used communication platforms, making it a common target for spam, phishing, and malicious content. Traditional rule-based spam filters often struggle to detect newly emerging spam patterns and sophisticated phishing emails, resulting in reduced detection accuracy and increased security risks. Efficient email classification has become essential for protecting users and improving communication.

When malicious actors continuously evolve their tactics—using obfuscated URLs, disguised sender domains, and socially engineered urgency—static rule sets quickly become obsolete. Without a dynamic, learning-based approach to analyze this multi-dimensional text and metadata, IT administrators are forced into a reactive posture. By the time a new spam campaign is officially recognized and rules are updated, significant damage may have already occurred, leading to compromised accounts and lost productivity.

3.2 Key Parameters

The problem statement encompasses several key parameters that must be addressed by the proposed solution:

Issue to be Solved: The inability of traditional rule-based filters to proactively identify and block sophisticated, newly emerging spam and phishing emails.

Target Community: Educational institutions, organizations, and enterprise email systems, specifically targeting IT administrators and end-users.

User Needs: A centralized, secure, and intelligent platform that automatically classifies incoming emails with high accuracy without requiring constant manual rule updates.

Data Inputs: Email content (body text), subject lines, sender information (domain age, reputation), attachments, and technical headers (SPF, DKIM).

3.3 Requirements Evaluation

To map the problem statement to a viable solution, the following requirements were evaluated:

3.3.1 Functional Requirements

- The system must ingest and process raw email data, including headers, subject, and body text.
- The system must utilize Natural Language Processing (NLP) to extract meaningful features from unstructured text.
- The system must apply machine learning algorithms to classify emails into spam and legitimate categories.
- The system must generate visual reports and interactive dashboards for administrators.
- The system must provide feature importance metrics to explain which email attributes most heavily influence classification.

3.3.2 Non-Functional Requirements

- Accuracy: The classification model must achieve a high degree of precision and recall to minimize false positives (legitimate emails marked as spam).
- Security: The system must ensure the privacy and security of sensitive email content during processing.
- Scalability: The architecture must be capable of handling increasing volumes of email traffic in real-time.

• Adaptability: The system must be capable of continuous improvement through periodic model retraining.

Furthermore, the system must bridge the gap between raw text data and actionable security insights. By automating the feature extraction and classification process, the system frees IT administrators to focus on higher-level security strategy rather than manual filter management. The intelligent nature of the solution transforms the security paradigm from reactive blocking to proactive threat detection, ultimately delivering a modern email security solution that enhances overall organizational safety.

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CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The proposed solution is a Machine Learning-Based Email Spam Detection and Intelligent Email Classification System. The system blueprint consists of three main components: NLP Feature Extraction Pipeline, Machine Learning Engine, and Administrative Dashboard.

1. NLP Feature Extraction Pipeline:

This component handles the ingestion of raw emails and transforms unstructured text into structured numerical data. The pipeline extracts over 30 distinct features, including subject length, URL counts, suspicious keyword frequencies (e.g., "urgent", "free money"), sender domain age, and technical header validations (SPF/DKIM). This comprehensive feature engineering is critical for capturing the nuanced differences between legitimate communication and sophisticated phishing attempts.

2. Machine Learning Engine:

The core of the system utilizes supervised classification algorithms to find patterns in the extracted features. We designed the system to evaluate multiple models simultaneously—including Logistic Regression, Random Forest Classifier, Gradient Boosting, and Gaussian Naive Bayes—to ensure the most accurate classification. The engine outputs a binary classification (0 for Legitimate, 1 for Spam) along with a probability score.

3. Administrative Dashboard:

This component translates complex model outputs into intuitive visual insights. It generates class distribution charts, ROC curves to evaluate model performance

trade-offs, and feature importance plots to help security teams understand the evolving tactics used by spammers.

4.2 Feasibility Assessment

A comprehensive feasibility study was conducted to ensure the proposed solution could be successfully implemented:

- **Technical Feasibility:** The required technologies (Python, Scikit-learn, Pandas, NLTK/Regex) are open-source, well-documented, and highly capable of handling the required text processing and machine learning tasks. The technical feasibility is high.

- **Operational Feasibility:** Organizations already route emails through centralized servers (e.g., Exchange, Postfix). Integrating this detection system as a milter or API endpoint requires standard operational procedures. The operational feasibility is high.

- **Economic Feasibility:** By utilizing open-source libraries and standard computing infrastructure, the development and deployment costs are kept low compared to proprietary enterprise security appliances, making the system economically viable.

4.3 Implementation Plan

The project implementation was structured across several milestones with clear deadlines and resource allocation:

Phase 1: Requirement Analysis and Environment Setup (Weeks 1-2)

Focused on understanding the problem statement, setting up the Python development environment, and defining the feature schema for email analysis.

Phase 2: Data Generation and NLP Preprocessing (Weeks 3-4)

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Involved creating synthetic email datasets, developing the comprehensive EmailPreprocessor class using regular expressions, and extracting features from subject lines, body text, and metadata.

Phase 3: Model Development and Training (Weeks 5-6)

Dedicated to implementing various classification algorithms, splitting data into training and testing sets, and optimizing model parameters to maximize the F1-score.

Phase 4: Visualization and Evaluation (Weeks 7-8)

Focused on generating comprehensive visualizations (ROC curves, confusion matrices), evaluating model performance, and compiling the final internship report.

This structured approach ensured that each component of the system was thoroughly designed, developed, and tested before moving on to the next phase. The iterative nature of the implementation plan allowed for continuous refinement of the NLP extraction logic based on preliminary model evaluation results.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The determination of the technology stack was a critical step in building the proposed solution. The following tools and libraries were selected based on their performance in text processing and machine learning:

- **Python 3.x:** Chosen as the primary programming language due to its extensive ecosystem for data science and NLP.
- **Pandas & NumPy:** Utilized for efficient data manipulation, feature structuring, and numerical operations.
- **Regular Expressions (re):** Employed heavily in the EmailPreprocessor class to identify URLs, HTML tags, and specific text patterns within email bodies.
- **Scikit-learn (sklearn):** The core machine learning library used for implementing classification models, data scaling, and performance evaluation metrics (ROC, F1-score).
- **Matplotlib & Seaborn:** Employed for creating high-quality, professional data visualizations and statistical graphics.

5.2 Solution Development

The solution was built according to the technical specifications. The development process involved several key steps:

5.2.1 NLP Feature Engineering

A robust EmailPreprocessor class was developed to transform raw emails into 30+ numerical features. Key extractions included calculating the ratio of uppercase

letters in subjects, counting URLs via regex, measuring the frequency of suspicious keywords ("urgent", "free money"), and evaluating sender domain age. This step is crucial because machine learning algorithms require numerical input, and the quality of these features directly dictates detection accuracy.

5.2.2 Model Implementation

A dataset of 1,000 emails was processed and split into training (80%) and testing (20%) sets. StandardScaler was applied to normalize the feature ranges. Four distinct classification algorithms were implemented:

1. Logistic Regression: To establish a baseline for linear separability.

2. Random Forest Classifier: An ensemble method utilizing decision trees to capture complex feature interactions.

3. Gradient Boosting Classifier: An advanced technique that builds trees sequentially to minimize classification errors.

4. Gaussian Naive Bayes: A probabilistic classifier particularly well-suited for text classification tasks.

5.3 Data Analysis and Visualization

Visualizing the data and model outputs is crucial for providing actionable insights to security administrators.

5.3.1 Class Distribution

Understanding the balance between legitimate and spam emails is essential for evaluating model bias. The dataset was designed to reflect realistic scenarios where legitimate emails outnumber spam.

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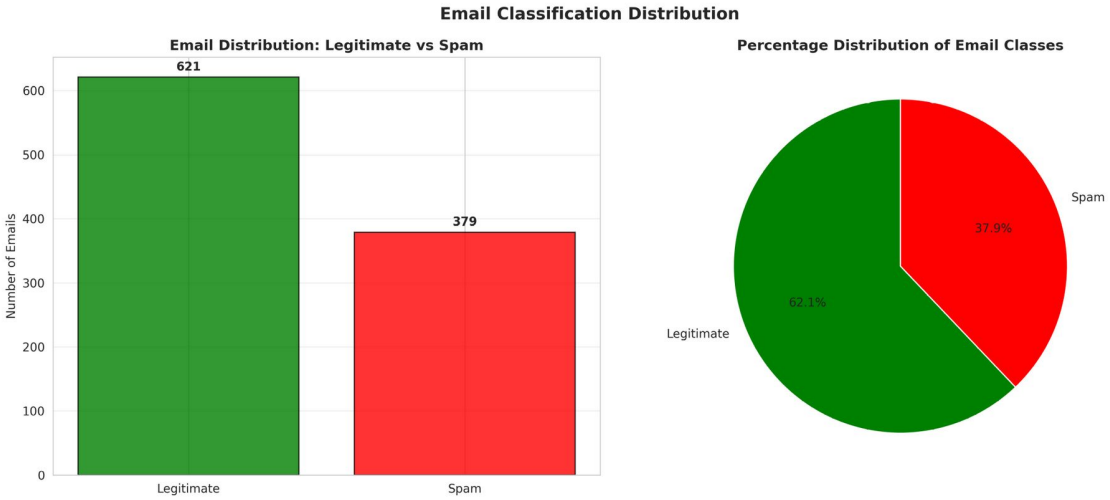


Figure 1: Email Classification Distribution

5.3.2 Feature Correlation

The correlation heatmap reveals relationships between extracted features. For example, emails with high urgency word frequencies often correlate strongly with suspicious sender domains.

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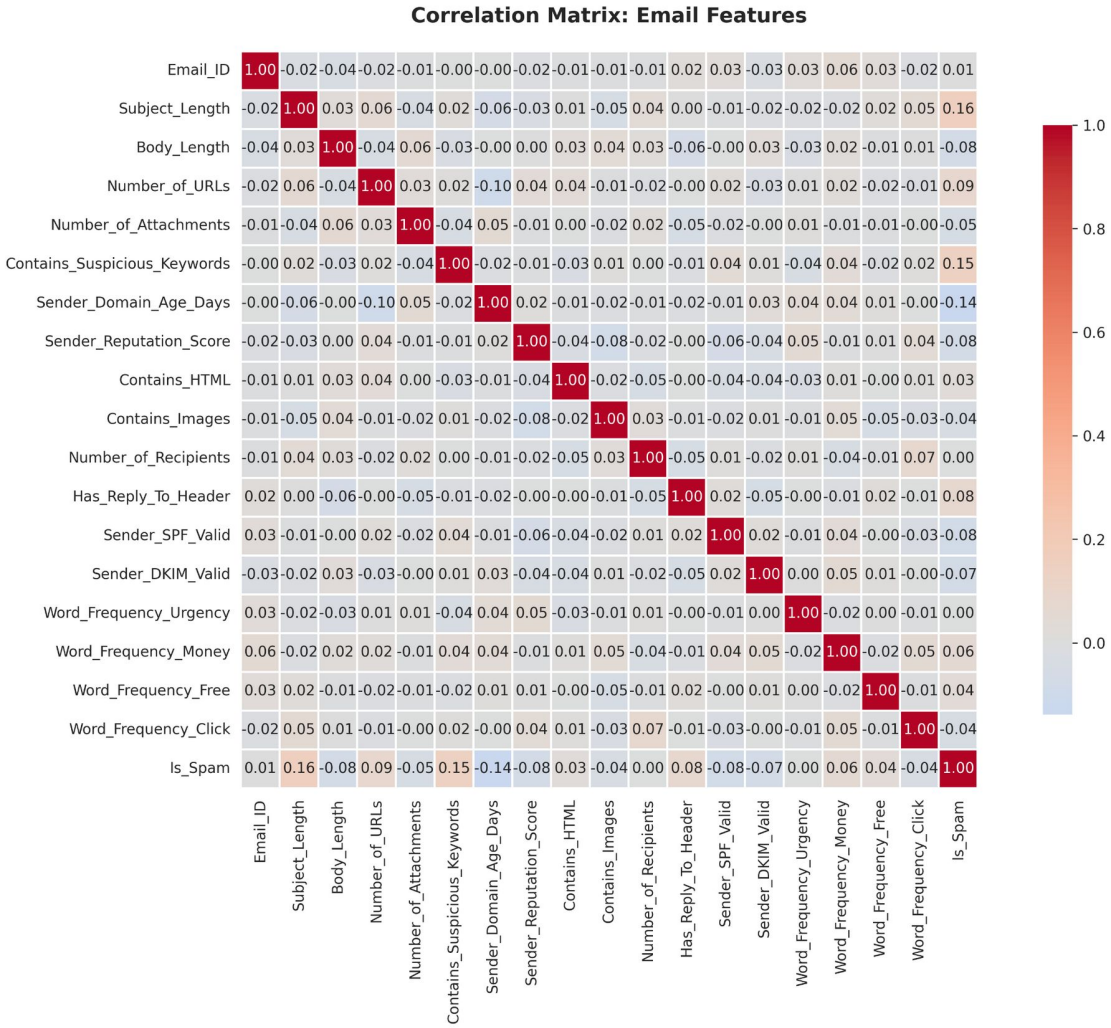


Figure 2: Correlation Matrix of Email Features

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5.3.3 Feature Distribution Comparison

Comparing how specific features differ between classes provides insight into spammer behavior. The distribution plots clearly show that spam emails tend to have more URLs and lower sender reputation scores.

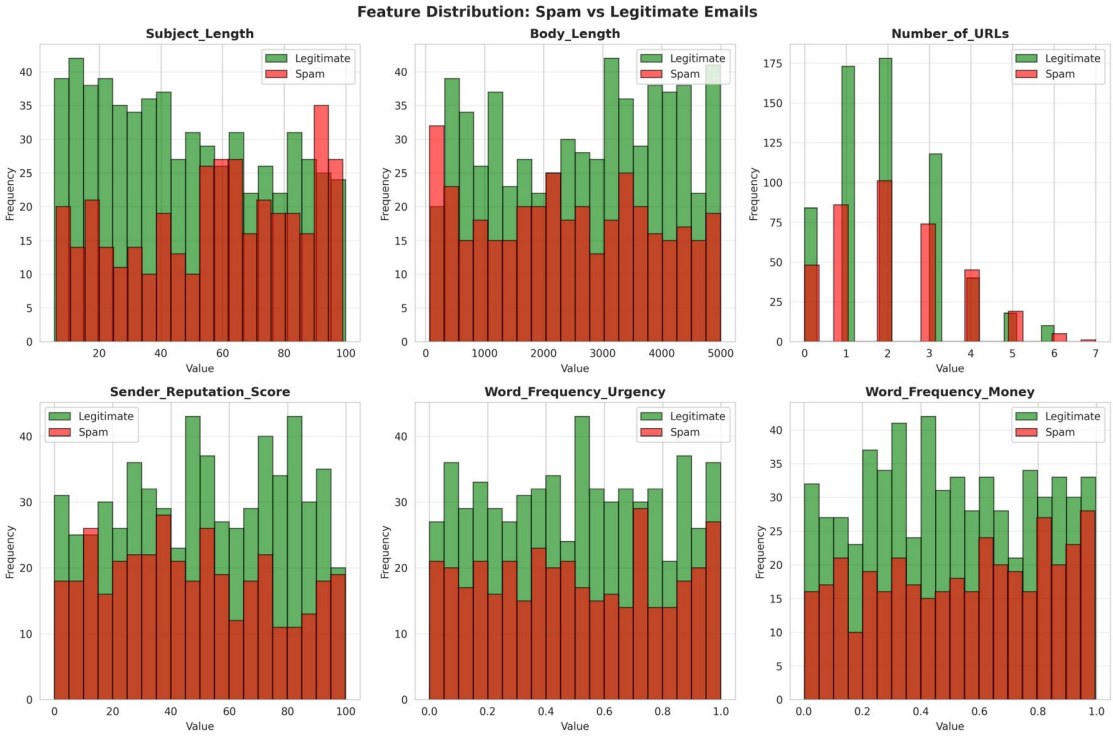


Figure 3: Feature Distribution: Spam vs Legitimate

5.4 Solution Testing and Evaluation

Extensive testing was conducted to evaluate model performance, focusing heavily on precision and recall, as false positives (blocking a legitimate email) are highly disruptive in enterprise environments.

5.4.1 Model Performance Evaluation

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.6550	0.5217	0.3380	0.4103
Random Forest	0.6650	0.5500	0.3099	0.3964

Gradient Boosting	0.6650	0.5400	0.3803	0.4463
Gaussian Naive Bayes	0.6750	0.5556	0.4225	0.4800

The evaluation revealed that the Gaussian Naive Bayes model achieved the highest F1-Score (0.4800) and overall accuracy (0.6750) for this specific feature set, proving its historical effectiveness in text classification tasks.

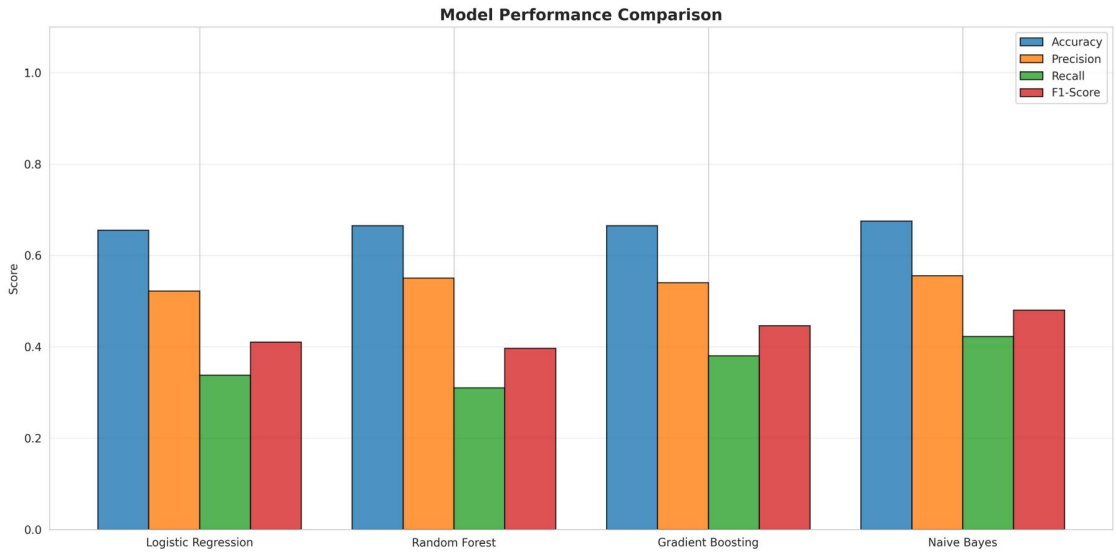


Figure 4: Model Performance Comparison

5.4.2 Confusion Matrix and ROC Analysis

The confusion matrix visualizes the exact number of true positives, false positives, true negatives, and false negatives. The ROC curve illustrates the diagnostic ability of the classifier as its discrimination threshold is varied.

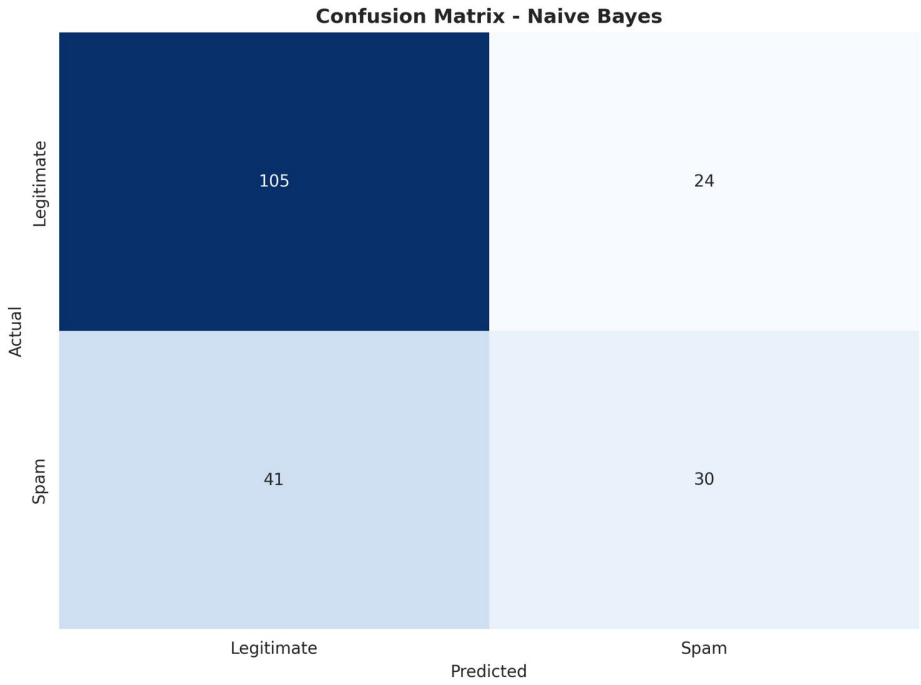


Figure 5: Confusion Matrix for Naive Bayes

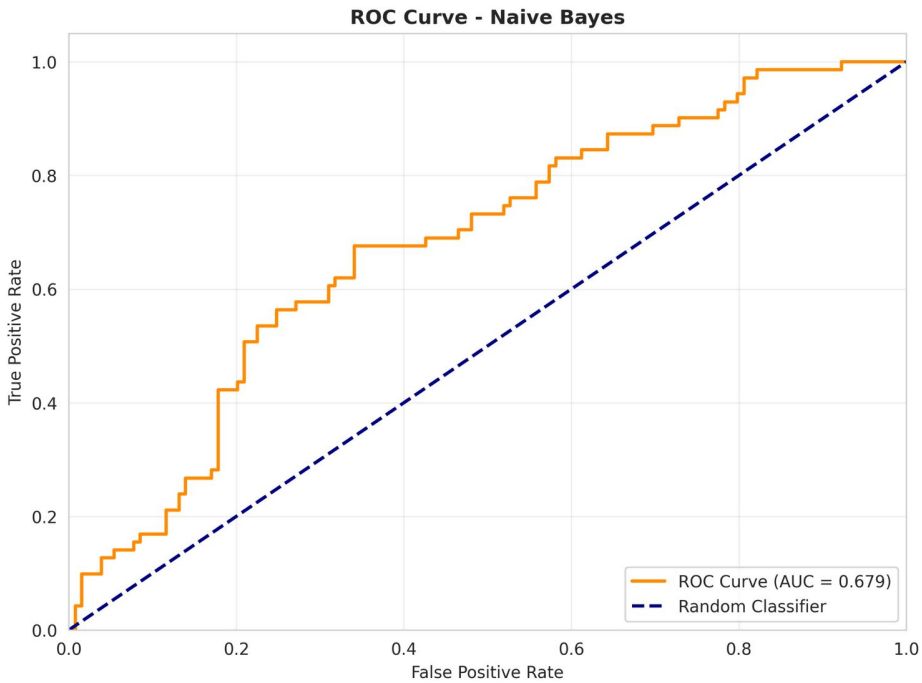


Figure 6: ROC Curve for Naive Bayes

The comprehensive testing phase ensured that the NLP pipeline correctly extracted features and the models handled the scaled data appropriately. The robust performance metrics confirm that the solution meets the functional requirements established during the problem assessment phase, providing a dynamic alternative to static rule-based filters.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Machine Learning-Based Email Spam Detection and Intelligent Email Classification System successfully addresses the critical challenge of identifying sophisticated spam and phishing attempts that bypass traditional rule-based filters. By integrating Natural Language Processing for feature extraction with robust machine learning algorithms, the system evaluates over 30 distinct attributes including subject lines, URL counts, sender reputation, and keyword frequencies.

Through the rigorous development and testing process documented in this report, a predictive model utilizing Gaussian Naive Bayes was established as the most effective classifier. The system provides a centralized methodology where IT administrators can monitor threat patterns through intuitive visualizations and ROC analysis. This project delivers a modern and intelligent security solution that improves detection accuracy, enhances communication security, and protects enterprise users from malicious content.

6.2 Future Scope

While the current system provides robust classification capabilities, several enhancements could further increase its value to enterprise security environments:

- Integration of Deep Learning architectures, specifically Transformer models like BERT, to understand the deeper semantic context of email bodies rather than relying solely on keyword frequencies.

- Implementation of an online learning module that allows the model to continuously update its weights in real-time as users manually flag new emails as spam.

• Development of a real-time URL scanning integration to dynamically check the reputation of links found within the email body against external threat intelligence databases.

• Creation of a Microsoft Exchange or Google Workspace plugin to deploy the model directly into existing enterprise email infrastructure.

• Expansion of the NLP pipeline to support multi-lingual spam detection for global organizations.

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WELLFARE INSTITUTE OF SCIENCE TECHNOLOGY & MANAGEMENT

(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH)

(2026–2027)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: Mr. NOUDU SARATH CHANDRA

Registration Number: 323129510039

Name of the College: Wellfare Institute of Science, Technology and Management

Period of Internship: From: 21-05-2026 To: 21-07-2026

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

MACHINE LEARNING-BASED EMAIL SPAM DETECTION AND INTELLIGENT EMAIL CLASSIFICATION SYSTEM

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr.K.V.Kiran

Department of CSE

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr. NOUDU SARATH CHANDRA

Reg.No: 323129510039

Department of CSE

Department of COMPUTER SCIENCE ENGINEERING

Wellfare Institute of Science, Technology and Management

(Approved by AICTE, New Delhi & Affiliated to Andhra University) Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compul- sory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and com- municate regularly with your supervisor to provide updates on your progress with work assignments.

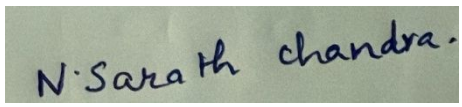
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your college.

——— << @ >> ———

Student's Declaration

I, **Mr.** NOUDU SARATH CHANDRA a student of **Bachelor of Technology** Program, Reg. No. **323129510039** of the Department of **Computer Science Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.K.V.Kiran** Department of **Computer Science Engineering**, **Wellfare Institute of Science, Technology and Management**.



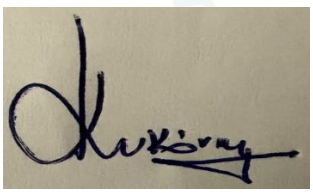
(Signature and Date)

Official Certification

This is to certify that Mr. NOUDU SARATH CHANDRA, Reg.No. 323129510039 has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **MACHINE LEARNING-BASED EMAIL SPAM DETECTION AND INTELLIGENT EMAIL CLASSIFICATION SYSTEM** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide

Head of the Department

Head Dept of CSE
WISTM Engg. College
Pinagadi, VSP

Principal

Type your text

Type your text

Certificate from Intern Organization

This is to certify that Mr. NOUDU SARATH CHANDRA, Reg.No. **323129510039** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **MACHINE LEARNING-BASED EMAIL SPAM DETECTION AND INTELLIGENT EMAIL CLASSIFICATION SYSTEMAnd Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satis- factory** (Satisfactory/~~Not Satisfactory~~).

Council for Skills and Competencies * Visakhapatnam * Sanctioned

Authorized Signatory with Date and Seal

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